LILY MORNINGSTAR

Professor of Cybersecurity and Digital Forensics, College of Southern Nevada

lily.morningstar@csn.edu • morningstar.lily@outlook.com • Las Vegas, NV

PROFESSIONAL SUMMARY

Cybersecurity educator, curriculum developer, and doctoral researcher with expertise in cybersecurity, digital forensics, artificial intelligence integration, experiential learning, and workforce-aligned curriculum design. Lead faculty member at the College of Southern Nevada with extensive experience developing academic programs, creating industry-aligned coursework, mentoring competitive cybersecurity teams, and contributing to national curriculum initiatives. Active contributor to cybersecurity education, AI workforce development, and emerging technology integration through ACM CCECC, CAE in Cybersecurity, NICE, and industry partnerships.

EDUCATION

Ph.D., Curriculum & Instruction — Interaction and Media Sciences (IMS), In Progress

University of Nevada, Las Vegas (UNLV) — Las Vegas, NV | Expected Graduation: Spring 2028

Research Interests: AI in education, cybersecurity education, gamification and experiential learning, accessibility and universal design for learning, student engagement and motivation, and learning analytics.

Ed.D., Curriculum & Instruction (27 credits completed; transferred to Ph.D. program, 2025)

University of Nevada, Las Vegas (UNLV) — Las Vegas, NV

Focus on learner-technology symbiotic interaction.

M.S., Cybersecurity and Digital Forensics, 2018

Utica University (formerly Utica College of Syracuse University) — Utica, NY

Focus on information security and computer crime investigation. **Capstone:** "Data Extraction from Solid State Drives."

M.S. / M.Ed. / B.S., Applied Mathematics and Informatics, 2007

North Caucasus Federal University (formerly Stavropol State University) — Stavropol, Russia

Areas: mathematical modeling, computer science, networking theory, and education. **Capstone:** "Data Transferring Over the Local Area Network Model."

SKILLS & COMPETENCIES

- Curriculum and instructional design
- Workforce-Aligned Program Design
- AI-Assisted Curriculum Development and Assessment Design
- Artificial Intelligence (AI) Integration in Higher Education
- AI Fluency, Responsible AI, and Human-AI Collaboration
- Gamification and Experiential Learning
- Instructional Technology & Online Learning Design
- Educational research design (quantitative, qualitative, and mixed methods)
- Learning Analytics and Educational Assessment
- Academic writing and scholarly communication
- Capture-the-Flag (CTF) Instruction and Competitive Coaching
- AI Literacy Instruction and Faculty Development

ACADEMIC APPOINTMENT & TEACHING EXPERIENCE

Professor of Cybersecurity and Digital Forensics 2019 – Present

College of Southern Nevada (CSN), Las Vegas, NV

Develop and teach undergraduate courses in cybersecurity and digital forensics; lead initiatives for inclusive learning environments leveraging emerging technologies. Tenured, 2023.

Courses Taught

- CIT 173 / CSEC 173 — Introduction to Linux (Spring 2020 – Present)
- CIT 174 / CSEC 174 — Linux System Administration (Fall 2025)
- CIT 217 / CSEC 217 — Security+ (Fall 2019 – Present)
- CIT 112 / CSEC 112 — Network+ (Fall 2019 – Present)
- CIT 274B / CSEC 281B — Ethical Hacking (Spring 2021 – Present)
- CSCO 230B — Security Operations Center Fundamentals (Fall 2023 – Present)
- CSEC 110B — Introduction to Digital Forensics, formerly CF 173 (Fall 2021 – Present)
- CSEC 290B — Security Capstone (Spring 2022)
- CSEC 226B — Compliance (Summer 2021)
- CSEC 114B — Introduction to Applied Network Forensics (Fall 2023)
- IS 101 — Introduction to Information Systems (Fall 2019; Spring 2020; 2024–2025)

Program Leadership & Course Development

- Lead curriculum development and modernization for the Cybersecurity – Network Security AAS, Cybersecurity – Digital Forensics AAS, and Cybersecurity – Digital Forensics CA programs.
- Led redesign of the Cybersecurity – Network Security AAS and Cybersecurity – Digital Forensics AAS/CA degrees, emphasizing hands-on labs, applied security skills, and alignment with EC-Council, CompTIA, and other industry standards.
- Developed specialization pathways within the Cybersecurity – Network Security AAS: Artificial Intelligence, Cloud Security, Offensive Security, Compliance, and Digital Forensics.
- Designed and launched new courses, including CSEC 100 (Cloud Security), CSEC 103 (IoT Security Essentials), CSEC 120/121/220/221 (Computer Hacking Forensics Investigator certification preparation), CSEC 181/281 (Certified Ethical Hacker Practical certification preparation), and CSEC 191 (Disaster Recovery & Business Continuity).
- Redesigned existing courses, including CSEC 110 (Digital Forensics Essentials, formerly CSEC 110B/CF 173), CSEC 287 (Threat Intelligence Essentials), CSEC 289 (Malware & Memory Forensics), and CIT 274 (Ethical Hacking Essentials), with updated learning outcomes, modernized labs, CTF-style activities, and alignment with current threat landscapes.
- Developed CIT 173 course map app aligned with UNLV curriculum for the CSEC 173 with a focus on CompTIA Linux+ certification preparation. https://github.com/LilMnstr-XAK3P/cit173-linux-map
- Developed autograded Linux labs for the course CIT 173 to assist students who do not have access to the computing resources that would allow them to complete labs and assist instructors with the proper grading process.
  - SCORM Linux Practice Labs https://github.com/LilMnstr-XAK3P/SCORM-Linux-Labs
  - SCORM Linux Blum's Linux+ Prep Practice Labs https://github.com/LilMnstr-XAK3P/SCORM-Blum-Linux-Plus
  - SCORM Linux Final Assessment Labs
- Developed Degree Advisor web & mobile app https://github.com/LilMnstr-XAK3P/degree-advisor
- Development and editing of the Cyber Center Web-page

NATIONAL CURRICULUM LEADERSHIP & WORKFORCE DEVELOPMENT

ACM CCECC CS-Transfer2026 Curriculum Development 2026 – Present

Contributor, ACM Committee for Computing Education in Community Colleges (CCECC)

- Selected contributor to a national computing curriculum initiative shaping CS-Transfer2026 guidance for two-year institutions, with contributions across Artificial Intelligence, Data Management, and Cybersecurity.
- Proposed Bloom's taxonomy–aligned revisions to learning outcomes, increasing cognitive rigor and assessment alignment.
- Contributed to cybersecurity domains including security engineering, digital forensics, cryptography, and AI in cybersecurity.
- Strengthened integration of Society, Ethics, and the Profession (SEP) and responsible computing principles within the cybersecurity curriculum.
- Provided input on emerging topics including adversarial machine learning and risk analysis.

Cybersecurity Curriculum & CTE Pathway Alignment 2026 – Present

College of Southern Nevada

- Contributing to program-level curriculum updates aligned with Fall 2026 implementation.
- Supporting alignment between high school CTE pathways and AAS cybersecurity degree programs, including course sequencing and articulation agreements.
- Improving transition pathways from secondary to postsecondary cybersecurity education and strengthening workforce alignment.

AP Program Curriculum Alignment & Assessment Contribution 2026

AP Reader / Contributor, Cybersecurity & Computing Education

- Evaluated cybersecurity course learning outcomes, curricula, and syllabi against AP-level academic standards for rigor and consistency with national benchmarks.
- Analyzed student performance data, comparing final examination scores with course grades, to evaluate assessment validity and grading alignment.
- Provided evidence-based recommendations supporting curriculum quality and assessment validity.

EC-Council AI Curriculum Training & Evaluation 2026

Contributor / Evaluator, EC-Council

- Completed professional development and curriculum evaluation activities for Artificial Intelligence Essentials (AIE), Certified AI Program Manager (CAIPM), Certified Offensive AI Security Professional (COASP), and Certified Responsible AI Governance & Ethics (CRAGE).
- Evaluated curriculum quality, workforce relevance, and integration of AI, security, governance, and ethics across certification pathways.
- Applied findings to inform the integration of AI and cybersecurity topics into institutional curriculum.

CAE in Cybersecurity Community 2025 – Present

Regular Participant

- Participate in national cybersecurity education initiatives focused on AI workforce development, AI Horizon, the National AI Research Resource (NAIRR), AI integration in cybersecurity education, and emerging educator resources.

PROFESSIONAL DEVELOPMENT & SCHOLARLY ACTIVITY

OpenAI – Higher Education training *(2026)*

- ***ChatGPT for Higer Education Faculty***
    - *Codex for Faculty and Researchers*
    - *Creating Workspace Agents for Higher Ed Faculty and Researchers*
    - *Building with Codex in Education*
- ***Codex for Everyday Use:*** Completed professional development on AI-assisted software development, workflow automation, and practical applications of Codex for coding, instructional design, and technical productivity.
- ***Builder Bootcamp: Evals:*** Completed advanced training on AI evaluation methodologies, including designing evaluation criteria, building repeatable evaluation workflows, using the OpenAI Evals API, analyzing failure patterns, and assessing AI system quality and reliability.
- ***Builder Bootcamp: RAG:*** Completed advanced training on Retrieval-Augmented Generation (RAG), including knowledge grounding, File Search, vector stores, retrieval workflows, and evaluation of grounded AI responses for reliability and trustworthiness.

Anthropic – Teaching AI Fluency *(Certificate of Completion, 2026)*

- Completed faculty certification in the Anthropic AI Fluency Framework for higher education.
- Studied the 4D AI Fluency Framework (Delegation, Description, Discernment, and Diligence) for responsible human-AI collaboration, authentic assessment, and AI-integrated curriculum design.
- Explored discipline-specific AI integration, AI-assisted assessment, authentic assignment design, and responsible AI implementation across higher education.

Black Hills Information Security (BHIS) Workshop
Prompt Engineering 201: The Context Stack *(Certificate of Completion, 2026)*

- Advanced training in context engineering, prompt design, AI workflow optimization, and reusable AI collaboration strategies for cybersecurity and education.

Microsoft Minecraft Education
Cyber Foundations for Educators *(Certificate of Completion, 2026)*

- Professional development in game-based cybersecurity education, experiential learning, and student engagement using Minecraft Education.

Doctoral Coursework Completed (33 credits)

- Curriculum theory and instructional design
- Educational research methods and statistics
- Learning sciences and interaction/media sciences
- Mixed methods and advanced research design

Innovative Teaching & Learning

- Design and implementation of gamified learning environments, including Capture-the-Flag (CTF)–based instruction.
- Development of original instructional materials grounded in learning theory.
- Faculty-mentored research preparation through Interaction and Media Sciences (IMS) coursework.

- Ongoing development of a research agenda focused on learning, motivation, and inclusion in technology-rich environments.
- Developed Capture-the-Flag (CTF)–based learning experiences and implemented experiential cybersecurity education models.
- Investigating gamification and immersive learning environments for cybersecurity instruction.

STUDENT SUCCESS & COMPETITIVE CYBERSECURITY COACHING

CSN CTF Squad Coach — National Cyber League (NCL)

Trained and mentored cybersecurity students in digital forensics, network security, OSINT, cryptography, incident response, and web security.

- Spring 2026 CSN Team placement: 44th/ nationally; College placement: 59th nationally.
- Competed against approximately 500 colleges & universities nationwide.

INDUSTRY ENGAGEMENT & WORKFORCE ALIGNMENT

Computing & Information Technology Advisory Committee, College of Southern Nevada

Collaborate with industry leaders representing healthcare, gaming, cloud computing, cybersecurity, and infrastructure services on:

- Curriculum review and modernization
- AI workforce readiness discussions
- AI governance and security considerations
- Student workforce preparation, internship, and employment pathways

PROFESSIONAL SERVICE

- Tenured Professor, College of Southern Nevada (2023 – Present)
- AI Force Initiative Ad-Hoc Committee Member (2024 – Present)
- AI Policy Development Contributor (2024 – Present)
- Modality Policy Development Contributor (2024 – Present)
- Student Authentication Policy Development Contributor (2024)
- Professional Advancement Committee Associate (2023 – 2025)
- Annual Degree Evaluation Committee Member (2022 – Present)
- Elected Faculty Senator, School of Applied Technology (2020 – 2024)

PROFESSIONAL AFFILIATIONS

- AP Reader – AP Cybersecurity Pilot Program, Contributor (2026)
- ACM Committee for Computing Education in Community Colleges (CCECC), Contributor (2026)
- CAE in Cybersecurity Community
- Women's Society of Cyberjutsu, WyCiS, ISSA

PRIOR PROFESSIONAL EXPERIENCE

Information Security Analyst 2017 – 2019

60out.com — Los Angeles, CA

Conducted risk assessments and safeguarded organizational data and system integrity.

- Conducted penetration testing and evaluated risks
- Performed information security compliance audits
- Delivered user security awareness education

- Implemented security best practices and policies

Owner-Operator 2012 – 2017

MATPELLIKA Group / MATPELLIKA Learning Center, LLC — Los Angeles, CA

Operated two businesses: MATPELLIKA Group provided technology support and services, including web application development, digital advertisement design, and photo and video editing. MATPELLIKA Learning Center, LLC provided licensed educational services for adults and children, including winter and summer break camps for elementary and middle school students.

VOLUNTEER EXPERIENCE

- Women's Society of Cyberjutsu — 2017 – Present
- Russian Orthodox Church of the Holy Transfiguration — 2012 – 2016